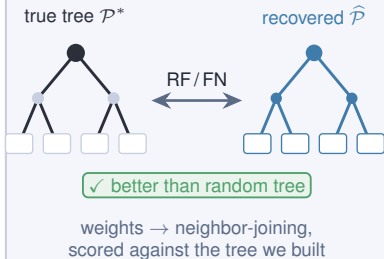


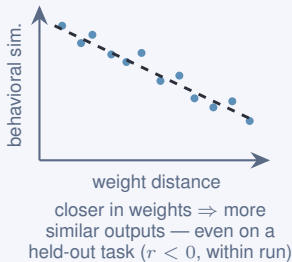
Fine-tuning leaves a path-dependent signature in weight space — a model *genealogy* that we validate against known trees, link to behavior, and recover from frontier models with no weight access.

A. Controlled recovery (*known truth*)



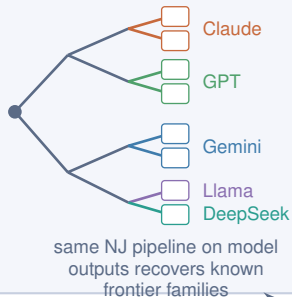
*full weight access,
known lineage*

B. Weights predict behavior



weight $\rightarrow$ behavior link

C. Frontier phylogeny (*no weights*)



*outputs only,
real frontier models*